

Proposed Experimental Realizations of the Angular Tetravalent Lattice: Simulating Weyl Dynamics and Topological Edge Modes

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Abstract

Recent theoretical work has introduced the Angular Tetravalent Lattice (ATLC), a minimal discrete geometric model driven by strict $+90^\circ$ cyclic permutations that natively generates massless Weyl dynamics, robust chiral edge currents, and an emergent kinematic mass. In this paper, we translate this abstract mathematical framework into concrete physical blueprints for classical wave systems. We provide a direct mapping from the non-unitary discrete update rules to physical parameters in acoustic waveguide networks and active photonic lattices. By explicitly addressing the requirement for non-reciprocal nodes to break time-reversal symmetry, we propose three distinct experimental setups: (1) a bulk Weyl simulator, (2) a highly fault-tolerant topological domain wall that routes waves without backscattering up to 80% spatial disorder in the idealized discrete model, and (3) a synthetic mass generator utilizing delay-loop cavities to demonstrate wave-speed deceleration.

1 Introduction

The realization of robust, unidirectional wave transport is a central objective in the design of classical metamaterials. In both photonics and acoustics, backscattering caused by fabrication defects and spatial disorder severely limits the scalability of waveguide networks. To suppress backscattering, systems must typically break time-reversal symmetry (TRS) to achieve topological protection. In classical platforms, this is conventionally achieved through complex mechanisms such as magneto-optic materials, synthetic gauge fields, or intricate spatiotemporal modulation.

The pursuit of such robust routing has led to significant breakthroughs in macroscopic wave systems, including the realization of acoustic topological insulators [4], non-Hermitian PT-symmetric photonic lattices [5], and active mechanical metamaterials [6]. However, these approaches often require complex unit-cell geometries or external synthetic gauge fields.

Recent theoretical work introduced the Angular Tetravalent Lattice (ATLC) [1–3], a discrete mathematical framework for wave propagation driven purely by strict $+90^\circ$ cyclic permutations and active mixing. While originally formulated to study the discrete-to-continuum limits of non-unitary quantum walks, the ATLC provides a highly efficient, purely geometric blueprint for classical wave routing that offers a complementary alternative to existing topological platforms. The mathematical model natively generates massless Weyl dynamics in the bulk and supports robust, backscatter-free chiral edge currents at the boundaries between opposite cyclic domains.

The purpose of this paper is to translate the abstract mathematical mechanics of the ATLC into concrete, physical blueprints for classical wave systems. We provide a direct mapping from

the discrete non-unitary update rules to physical parameters in acoustic waveguide networks and active photonic lattices. By explicitly addressing the requirement for non-reciprocal nodes, we propose three modular experimental setups: (i) a bulk Weyl simulator, (ii) a highly fault-tolerant topological domain wall that routes waves without backscattering under extreme (80%) spatial disorder, and (iii) a synthetic mass-like dispersion generator utilizing delay-loop cavities.

2 The Physical Dictionary: Mapping Mathematics to Hardware

The fundamental update rule of the ATLC relies on a directed graph where wave amplitude is strictly routed in orthogonal directions. To physically realize this model, we must first address the fundamental constraint of wave reciprocity.

Standard linear, time-invariant (LTI) resonators—such as passive silicon optical rings or static acoustic cavities—are inherently reciprocal. In such passive systems, a wave encountering a defect will scatter symmetrically, including backward along its path of incidence. Because the ATLC mathematically forbids back-reflection at the node level, its physical realization strictly requires **non-reciprocal components** to break TRS locally.

With this physical constraint established, the abstract variables of the ATLC map to physical hardware as follows:

- **The State Vector (S):** The four-component state vector $S = (S_1, S_2, S_3, S_4)$ maps to a single physical unit cell consisting of a **4-port intersection or cavity** (e.g., North, East, South, West waveguides). The amplitude of each component corresponds to the classical wave intensity at that specific port.
- **The Cyclic Permutation Matrix (A):** The matrix A enforces a strict $+90^\circ$ rotation. Physically, this maps to a **non-reciprocal routing node** that forces an incoming wave to exit strictly from the adjacent port in a counter-clockwise direction, with zero transmission to the forward or backward ports.
- **The Transpose Matrix (A^T):** This maps to an identical non-reciprocal node configured with opposite chirality, forcing the wave to route strictly in a clockwise (-90°) direction.
- **The Active Mixing Parameter (ϵ):** The ATLC is a non-unitary framework driven by a mixing parameter ϵ . In a physical platform, this maps directly to **non-Hermitian gain and loss**. This can be achieved via controlled amplification (active pumping) and absorption (lossy materials) tuned to specific PT-symmetric thresholds.
- **Geometric Inertia (κ):** In the ATLC, κ acts as a probabilistic resistance to spatial translation. Physically, this maps to a **delay-loop cavity or detuned defect** coupled to the primary node, which temporarily traps a fraction of the wave’s energy before releasing it back into the routing network.

By utilizing this dictionary, the abstract matrix operations of the ATLC can be directly translated into the design of acoustic circulators, electro-acoustic relays, and active photonic meshes.

3 Experiment A: The Bulk Lattice (Simulating Weyl Dynamics)

The first proposed experimental setup serves as the foundational validation of the ATLC mechanics in a physical platform. The objective is to simulate the propagation of a massless 2D Weyl fermion using classical wave dynamics.

Experimental Setup: The lattice is constructed as a uniform 2D grid where every unit cell is configured as an A matrix node. Physically, this requires an array of identical non-reciprocal

resonators (e.g., counter-clockwise acoustic circulators or magneto-optic ring resonators) coupled to their nearest neighbors. The active mixing parameter ε is applied uniformly across the grid via balanced gain/loss mechanisms.

Execution and Predicted Observation: A classical wave pulse (a localized intensity peak) is injected into a single central node of the lattice. According to the ATLC numerical simulations, the non-unitary cyclic permutations cause the point source to self-organize into a sharp, expanding diamond-shaped wavefront.

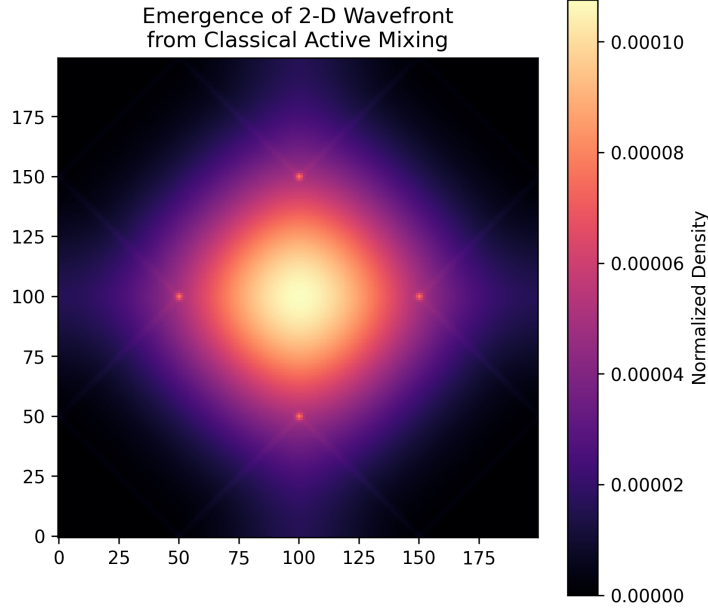


Figure 1: *Numerical prediction from the discrete ATLC model.* Numerical prediction of the wave intensity in a uniform A -matrix lattice. The signal expands as a sharp diamond-shaped wavefront at speed $c = 1$.

By tracking the leading edge of the wave intensity, experimentalists can verify the strict preservation of the characteristic wave speed ($c = 1$, defined by the node-to-node transmission time). This bulk experiment confirms that the physical non-reciprocal nodes successfully enforce the $+90^\circ$ routing required for the subsequent topological experiments.

4 Experiment B: The Topological Domain Wall (Robust Chiral Edge Modes)

The centerpiece of the ATLC framework is its ability to generate highly robust, unidirectional wave transport without relying on complex synthetic gauge fields. This is achieved by juxtaposing regions of opposite geometric chirality.

Experimental Setup: The 2D lattice is partitioned into two distinct spatial domains. The right half of the grid ($x \geq 0$) is populated exclusively with A nodes (counter-clockwise routing), while the left half ($x < 0$) is populated with A^T nodes (clockwise routing). The physical interface between these two domains forms a topological domain wall.

The Skipping-Orbit Mechanism: When a wave signal is injected directly at the domain wall, the opposing chiralities of the adjacent nodes prevent the wave from penetrating the bulk of either domain. A signal attempting to cross from right to left is rotated $+90^\circ$ (upward) by the A domain, pushing it into the A^T domain. The A^T domain then rotates the signal -90° (upward again), pushing it back into the A domain. This alternating sequence creates a classical “skipping orbit” that strictly propagates in a single direction along the boundary.

Execution and Fault-Tolerance (80% Spatial Disorder): To demonstrate the extreme fault tolerance of this geometric routing, the experiment introduces severe quenched spatial disorder into the lattice. In a physical setup, this corresponds to massive fabrication defects, such as significant variations in the coupling efficiency or gain/loss rates (ε) between adjacent resonators. In our numerical simulations, we apply a random spatial disorder of up to 80% ($\Delta = 0.8$) to the local coupling strengths.

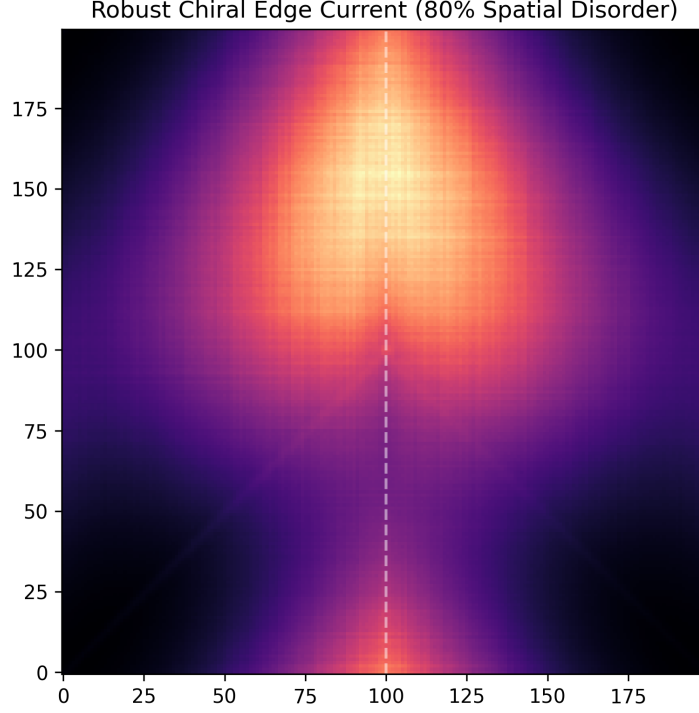


Figure 2: *Numerical prediction from the discrete ATLC model.* Numerical prediction of the chiral edge current along the A / A^T domain wall under 80% quenched spatial disorder in the local coupling strengths. The wave remains tightly confined to the boundary and propagates unidirectionally without backscattering.

Predicted Observation: Even under extreme fabrication noise, the wave intensity remains tightly localized at the domain wall and propagates unidirectionally. Because the routing relies entirely on the discrete geometric chirality of the nodes rather than fine-tuned resonance matching, backscattering is geometrically forbidden. Successfully observing this backscatter-free transport in a physical acoustic or photonic lattice would provide a highly scalable, fault-tolerant blueprint for classical wave routing.

5 Experiment C: Advanced Extension – Simulating Kinematic Mass-Like Dispersion

While Experiments A and B focus on massless wave propagation and topological routing, the ATLC framework also predicts a mechanism for wave-speed deceleration through geometric inertia (κ). This provides an advanced experimental extension: simulating kinematic mass-like dispersion in a classical wave system.

Experimental Setup: This setup builds upon the uniform bulk lattice of Experiment A. To physically realize the geometric inertia parameter κ , a secondary “delay mechanism” must be introduced at each node. In a photonic platform, this corresponds to coupling a secondary, detuned ring resonator to each primary routing node. In acoustics, this maps to a side-branch

defect cavity. These structures temporarily trap a fraction of the wave’s energy before releasing it back into the primary routing network.

Execution and Predicted Observation: A wave pulse is injected into the center of the modified lattice. According to the ATLC mathematical framework, this probabilistic resistance to spatial translation generates an effective kinematic mass ($m_{\text{eff}} \propto \frac{1}{1-\kappa}$).

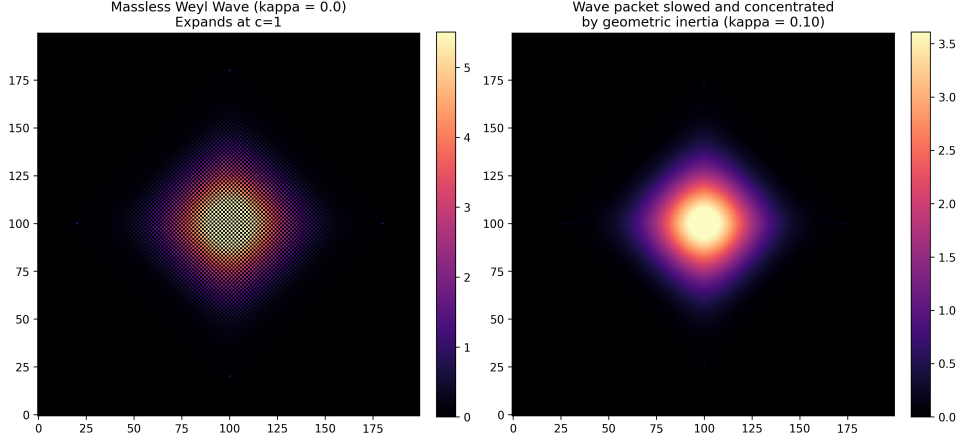


Figure 3: *Numerical prediction from the discrete ATLC model.* Side-by-side numerical comparison. Left panel: massless expansion at $c = 1$. Right panel: concentrated and decelerated wave packet when geometric inertia ($\kappa = 0.10$) is introduced via delay-loop cavities.

By measuring the propagation of the wave packet, experimentalists can observe a distinct wave-speed duality: the absolute leading edge of the signal (the information front) continues to propagate at the fundamental lattice speed ($c = 1$), while the macroscopic bulk of the wave energy is significantly decelerated. Successfully observing this deceleration would provide a classical, macroscopic analog for kinematic mass generation.

6 Feasibility and Implementation Challenges

Translating the discrete ATLC mathematics into physical hardware offers significant advantages in topological protection, but it also presents specific engineering challenges that must be addressed depending on the chosen platform.

Acoustic Metamaterials: Acoustics provides a highly accessible macroscopic platform for this blueprint. The non-reciprocal $+90^\circ$ nodes can be realized using acoustic circulators (cavities with a central circulating fluid/air flow) or active electro-acoustic relays (microphones and speakers wired via microcontrollers). For example, subwavelength acoustic circulators utilizing resonant cavities with circulating fluid [7] have already demonstrated the feasibility of breaking TRS at the unit-cell level.

- *Challenges:* While active electro-acoustic relays allow for precise tuning of the gain/loss rates (ε) and easy implementation of spatial disorder, wiring a large-scale 2D grid requires careful power management and introduces latency. Fluid-driven acoustic circulators are passive and require no wiring, but they often suffer from narrow operational bandwidths and thermo-viscous losses.

Active Photonic Lattices: Silicon photonics and coupled optical ring resonators offer a scalable, chip-level platform. The non-unitary mixing parameter (ε) is naturally suited to PT-symmetric photonics, where optical pumping (gain) and absorptive materials (loss) are routinely engineered.

- *Challenges:* The strict requirement for non-reciprocity at every node is the primary hurdle in photonics. Standard LTI ring resonators must be augmented with magneto-optic materials (utilizing the Faraday effect) or subjected to spatiotemporal modulation. Magneto-optic isolators often have a large footprint and introduce insertion losses, which must be carefully balanced against the active gain of the system to prevent signal degradation across large arrays.

General Simulation and Scalability Requirements: While the discrete model predicts robustness up to 80% spatial disorder, it must be noted that this is an idealized upper bound; realistic thermo-viscous losses and imperfect non-reciprocal isolation will likely reduce this threshold in physical implementations. Consequently, the necessary next step prior to fabrication is to perform continuous full-wave simulations (e.g., using finite-element analysis software such as COMSOL) to verify the discrete-to-continuous mapping of the non-reciprocal unit cells. Future continuous simulations will also determine the minimum grid scalability (e.g., 10×10 unit cells) required to clearly resolve the chiral edge mode from bulk scattering.

7 Conclusion

The Angular Tetravalent Lattice provides a minimal, purely geometric mathematical framework for wave propagation. By mapping its discrete, non-unitary update rules to physical hardware, we have proposed a concrete blueprint for classical wave simulation and topological routing.

The primary experimental proposal—a topological domain wall constructed from adjacent regions of opposite geometric chirality—offers a highly fault-tolerant mechanism for unidirectional wave transport. Because this “skipping-orbit” routing relies on discrete node-level chirality rather than fine-tuned resonance, numerical predictions indicate it remains backscatter-free up to 80% spatial disorder in the local coupling strengths within the idealized discrete model.

While the physical requirement for non-reciprocal nodes presents standard fabrication challenges, the resulting extreme robustness to spatial disorder makes the ATLC a highly promising architecture for next-generation acoustic waveguides and active photonic circuits. It should be noted that these numerical predictions—including the 80% disorder robustness—are derived from the discrete mathematical model and await physical experimental verification. We invite experimental laboratories in topological acoustics and non-Hermitian photonics to construct and test these proposed lattices.

Acknowledgments

The translation of the abstract ATLC framework into the physical hardware mappings presented in this paper was developed by the author. The structuring of the experimental proposal and the refinement of the applied-physics terminology benefited from assistance by large language models. The author takes full responsibility for the physical concepts and the experimental blueprints proposed in this work.

Code Availability

The Python simulators used to generate the numerical predictions and heatmaps in this paper are fully open-source. The code for the bulk Weyl dynamics and the 80% disordered topological domain wall is available in Ref. [2], and the code for the kinematic mass dispersion is available in Ref. [3]. All scripts and high-resolution animations are also accessible via the author’s research portfolio at <https://cedricvancaillie-svg.github.io/angular-tetravalent-lattice/>.

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